

RF Design Course 2026

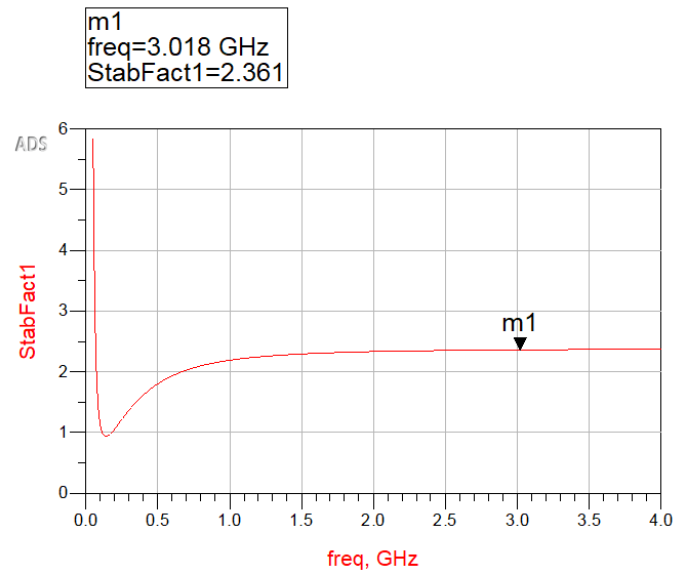
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July 13, 2026

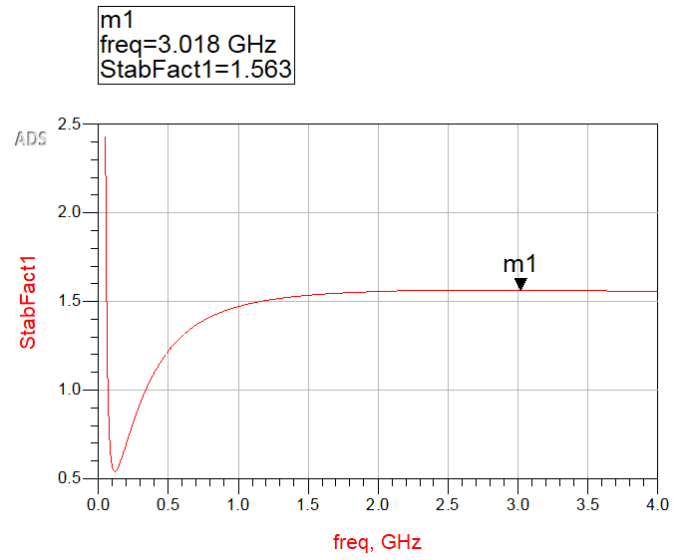
1 Task 17

1.1 Number 4

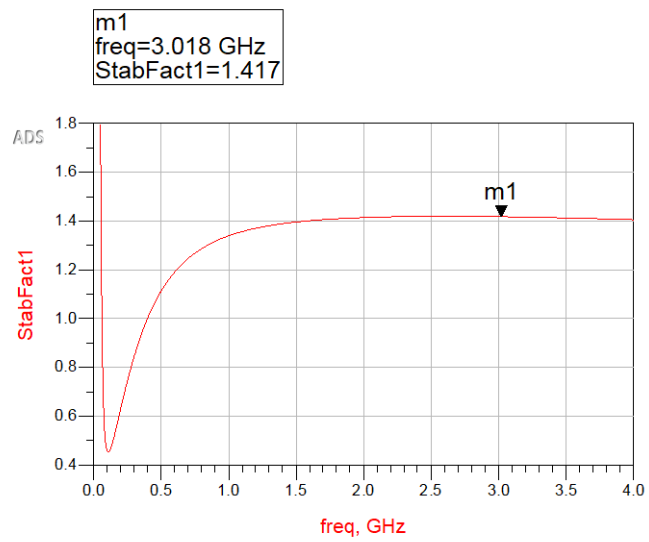
1.1.1 Vdc=0.45V



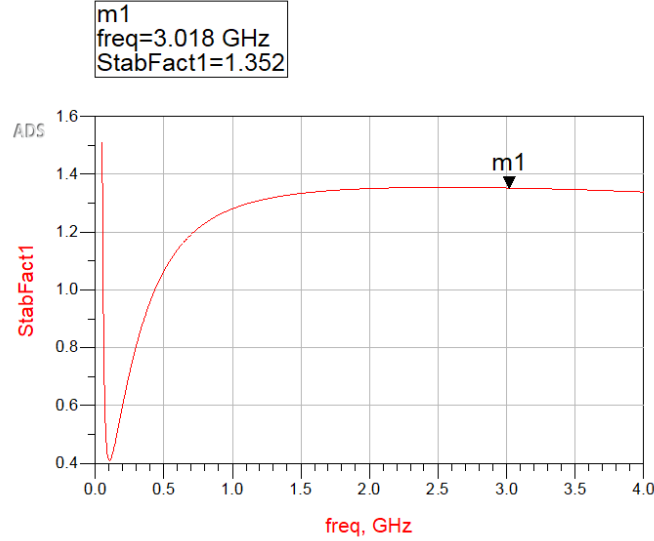
1.1.2 Vdc=1.05V



1.1.3 Vdc=1.55V



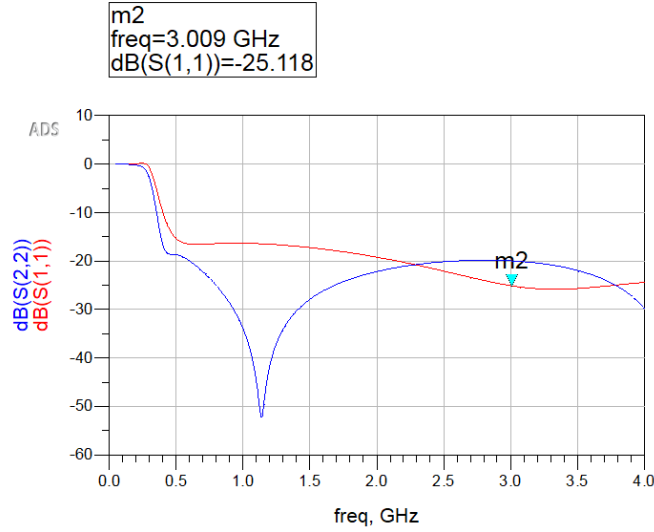
1.1.4 Vdc=2.05V



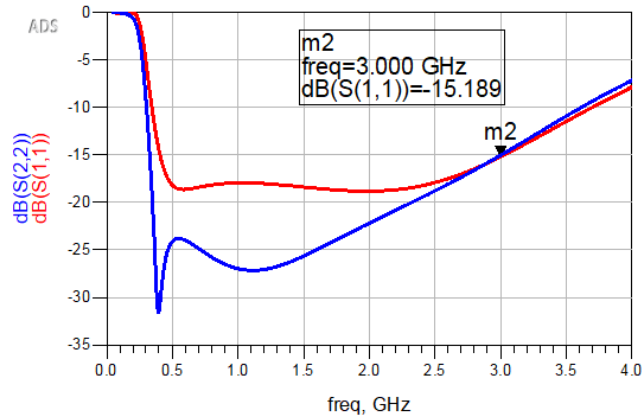
- When the frequency is swept from 0.05 GHz to 4 GHz, the stability factor exhibits a sharp decline at lower frequencies below 1 GHz, dropping below unity in the 100 MHz range.
- This critical degradation indicates that the low-frequency region is highly susceptible to unconditional instability, creating a potential operating zone where unintended oscillations can easily be triggered.
- Furthermore, increasing the bias voltage from 0.45 V to 2.05 V significantly accelerates this drop, demonstrating that higher supply voltages increase the device's gain and transconductance, which severely worsens the stability margin.
- POSSIBILITY : If the packaging environment drives the amplifier into this unstable regime (when *StabilityFactor* < 1), the circuit may experience parasitic high-frequency oscillations that distort the desired signal or, in a worst-case scenario, draw excessive current that leads to permanent thermal destructive breakdown of the MMIC.

1.2 Number 5

The simulation is done without the package



Now the same simulation is done with the package



- Comparing the two states at the 3 GHz marker shows that adding the QFN package degrades the input matching (S_{11}) from -25.118 dB to -15.189 dB.
- This upward shift in both reflection coefficients across the band indicates a noticeable impedance mismatch introduced by the parasitic capacitances and inductances of the package leads and bond pads.
- As the frequency approaches 0 Hz (DC), both curves converge toward 0 dB, demonstrating that the matching network naturally loses its impedance-matching capabilities at extremely low frequencies regardless of the packaging state.

1.3 Number 6

QFN Package Resonance Analysis

The package dimensions are:

$$x = 6.1971 \text{ mm}, \quad y = 9.87552 \text{ mm}, \quad z = 5.31884 \text{ mm}$$

The quarter wavelength condition is used to determine the resonance frequencies:

$$\frac{\lambda_0}{4} = L$$

where the free-space wavelength is:

$$\lambda_0 = \frac{c}{f}$$

with $c = 3 \times 10^8 \text{ m/s}$.

The calculated wavelengths and corresponding frequencies are:

Direction	$\lambda_0(\text{mm})$	$\lambda_0/4(\text{mm})$	$f(\text{GHz})$
x	24.7884	6.1971	12.10
y	39.5021	9.8755	7.59
z	21.2754	5.3188	14.10

At these frequencies, the package dimensions become electrically large and the QFN package can behave as a cavity resonator, similar to a musical instrument. This can introduce standing waves, unwanted coupling between input and output, impedance mismatch, and possible degradation of RF performance.

For a plastic mold with dielectric constant:

$$\epsilon_r = 2$$

the wavelength reduces by:

$$\lambda_d = \frac{\lambda_0}{\sqrt{\epsilon_r}}$$

and the resonance frequencies shift to:

$$f_d = \frac{f_0}{\sqrt{2}}$$

giving:

Direction	$f_d(\text{GHz})$
x	8.56
y	5.37
z	9.97

Thus, the plastic mold lowers the package resonance frequencies, making cavity effects occur at lower frequencies.

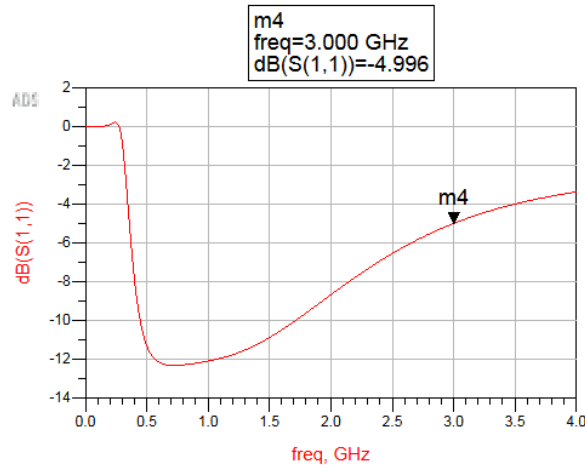
1.4 Number 7

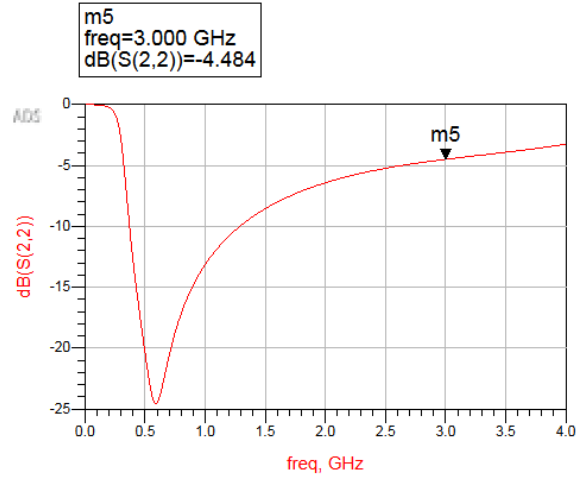
A typical characteristic impedance attributed to standard microstrip pads on a high-frequency PCB or package substrate is $50\ \Omega$.

- When analyzing the connecting bond pads as short Microstripline (MSL) transmission line segments, they are generally designed to approximate a characteristic impedance (Z_0) of $50\ \Omega$ to match the reference impedance of the internal MMIC ports and external test equipment.
- In practice, the realistic range of characteristic impedances that can be successfully manufactured with reasonable pad physical dimensions spans from approximately $20\ \Omega$ to $100\ \Omega$.
- Realizing impedances outside this window becomes highly impractical because lowering the impedance below $20\ \Omega$ requires excessively wide pads that introduce disruptive parasitic shunt capacitances, whereas exceeding $100\ \Omega$ requires narrow conductor lines that are bounded by manufacturing tolerances and carry high series resistance.

1.5 Number 8

The bond wire is emulated with the help of the inductor. The simulation is shown as below:





- Incorporating the 4-turn inductor at the input to model the bondwire parasitic degrades the input reflection coefficient (S_{11}) at the 3 GHz marker by increasing it from -25dB to the value up toward 0 dB, introducing an additional mismatch penalty measured in decibels.
- Similarly, adding the identical parasitic inductor to the output port causes a matching deviation in S_{22} , shifting the resonance curves away from the intended design center frequency.
- This upward shift in reflection magnitude across the operating band indicates that the inductive parasitic behavior of physical wire bonds significantly perturbs the carefully optimized impedance match between the internal MMIC pads and the external package interface.
- For a practical packaged device, this degradation implies a reduction in power transfer efficiency, an increased risk of signal reflections, and a potential reduction in the overall operating bandwidth of the Low Noise Amplifier (LNA).

2 Task18

2.1 Number3

- Important variations across a 12-inch CMOS wafer include epitaxial or oxide-layer thickness, reactor temperature, lithographic dimensions, and etching depth.
- These process variations cause spatial changes in transistor parameters such as threshold voltage, transconductance, resistance, and capacitance.

- They must be considered because chips fabricated at different wafer positions can exhibit different RF performance.

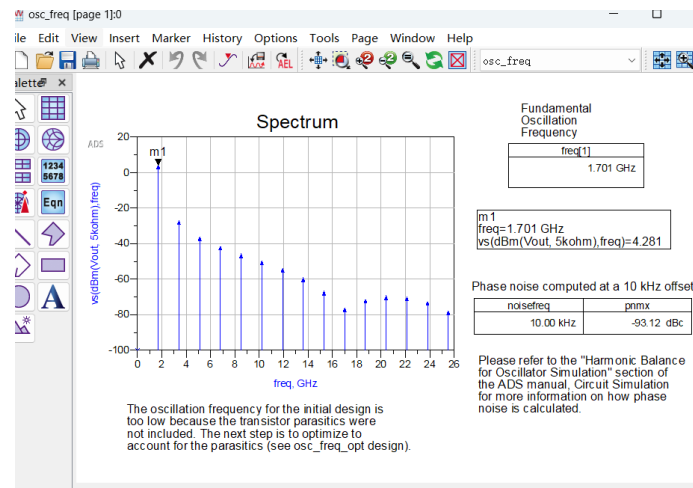
2.2 Number4

- Low yield wastes wafers, chemicals, energy, and processing time.
- Higher yield reduces scrap and lowers the environmental cost per working chip.
- It also avoids unnecessary packaging, testing, transport, and disposal of defective parts.

2.3 Number5

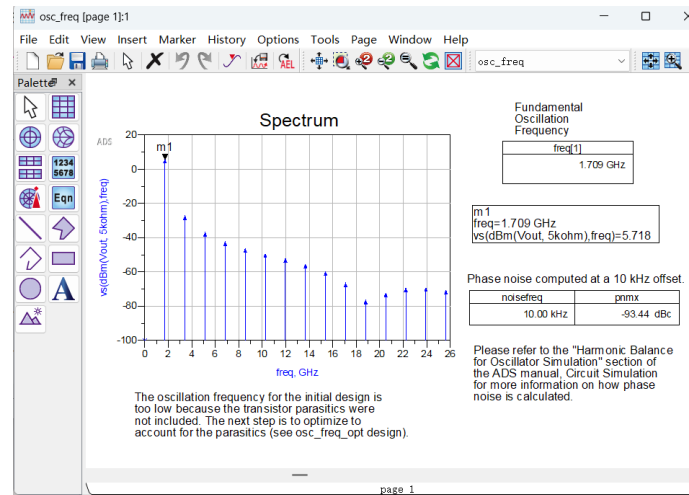
- An approximately independent example is the oscillator-frequency limit and the maximum chip-area limit, since they are determined mainly by different design factors.
- A dependent example is output power and DC power consumption, because increasing the transistor bias current can increase both quantities.

2.4 Number6



- The default simulation gives an oscillation frequency of 1.701 GHz, which is 299 MHz below the targeted 2 GHz.
- The initial component values were calculated without fully accounting for the transistor parasitics included in the simulation model.

- These parasitics increase the effective resonator capacitance and therefore reduce the oscillation frequency.



- After setting the transistor parasitic resistances R_b , R_c , and R_e to zero, the frequency increases to 1.709 GHz and is therefore 8 MHz closer to the target.

2.5 Number7

- Too high oscillator phase noise broadens the carrier spectrum.
- This can increase adjacent-channel interference and reduce signal quality in a communication system.
- It can also increase bit-error rate and reduce receiver sensitivity.

2.6 Numebr9

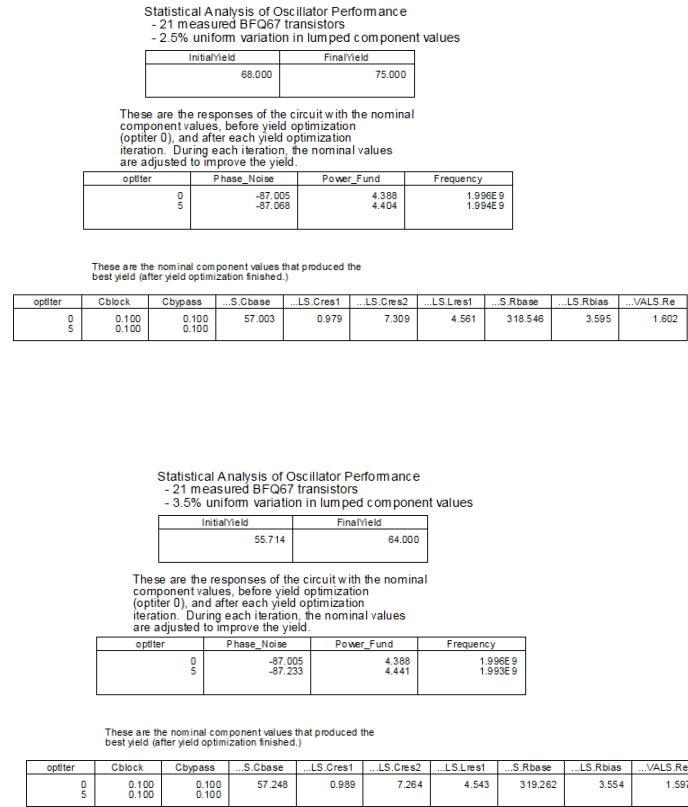
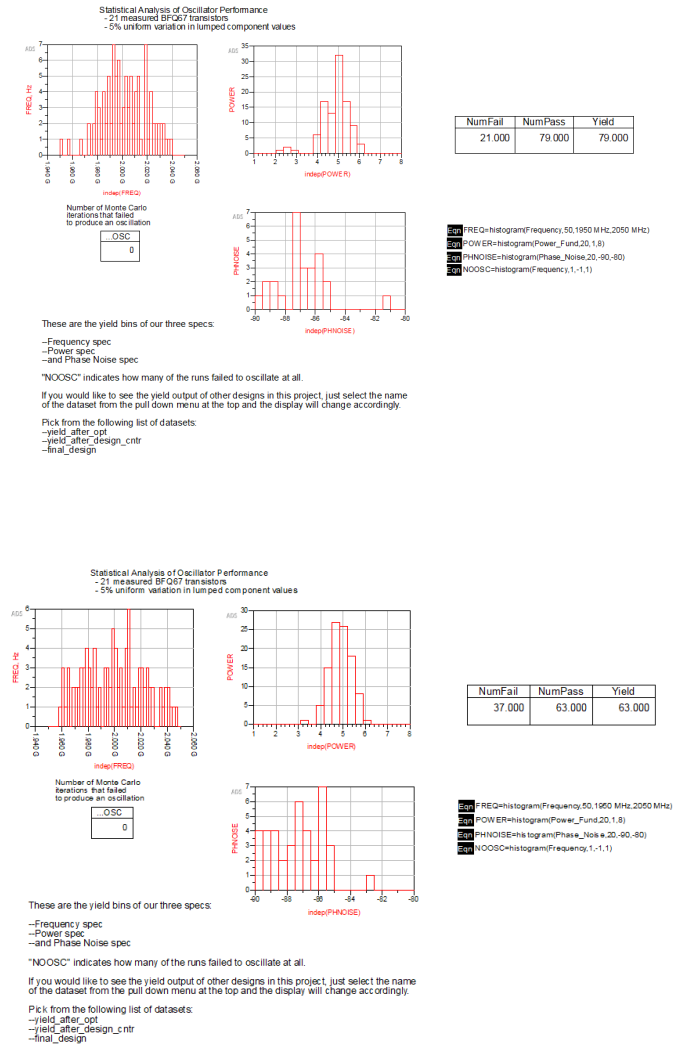


Figure 1: Enter Caption

- The default $\pm 5\%$ setting gives an initial RF yield of 34.000% and a final yield of 48.571%.
- Reducing the component variation to $\pm 2.5\%$ increases the initial and final yields to 68.000% and 75.000%, respectively, whereas $\pm 3.5\%$ gives 55.714% and 64%.
- Therefore, tighter control of component variations significantly improves RF yield, with the $\pm 2.5\%$ setting producing the best result.

2.7 Number11



- The final design gives RF yields of 79%, and 63% for the $\pm 2.5\%$, and $\pm 3.5\%$ variations.
- Repeated simulations produced some variation because only 100 Monte Carlo samples were used.
- Reducing individual component variations does not necessarily produce a proportional yield improvement because transistor variations and parameter interactions also affect whether all specifications are satisfied.

- Therefore, yield must be treated as a system-level target based on the simultaneous fulfillment of all specifications rather than optimized independently for individual components.